SMART TRANSPORTATION NETWORK DESIGN

Graph Theory & Route Optimization for Large-Scale Urban Infrastructure Modeling

PHASE 1: SYNTHETIC CITY MAPPING

PROBLEM STATEMENT & SCOPE

The Objective

Develop a high-fidelity synthetic state network using graph-theoretic principles. We model a newly developed region consisting of 500 interconnected city nodes to analyze logistical efficiency and connectivity.

Industry Context

Applications in modern urban planning, autonomous vehicle logistics, delivery systems, and global mapping services like Google Maps or FedEx logistics engines.

Problem Setup

Generate a virtual map consisting of 500 cities represented by IDs from 0 to 499. For simplicity, city IDs will be used instead of real city names. In real-world applications, these IDs could be mapped to actual locations. Cities should be connected randomly to form a transportation network. Each connection (edge) must have a distance between 5 km and 50 km, where 1 unit represents 1 km.

The generated network should be stored using an adjacency list representation.

TASK A: NETWORK ARCHITECTURE

Generate all city connections and assign random distances between connected cities(explore rand() and srand() function for random generation). Store the network as an adjacency list and display a summary of generated nodes and edges.

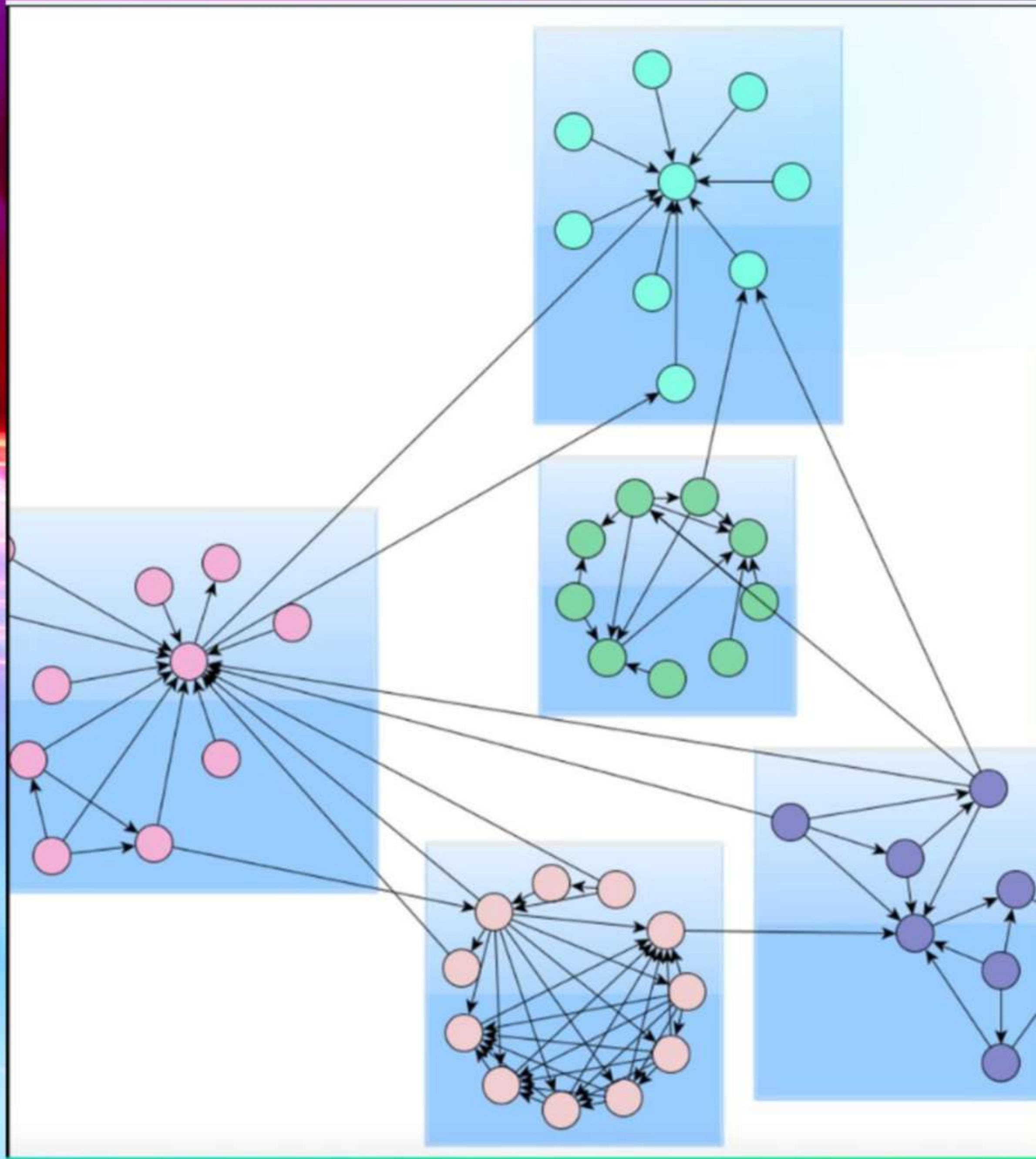
TASK B: CONNECTIVITY ANALYSIS

Using DFS or BFS, determine the largest group of cities that are connected through some path.

Report: Total number of cities in the largest connected component. List of cities belonging to that component.

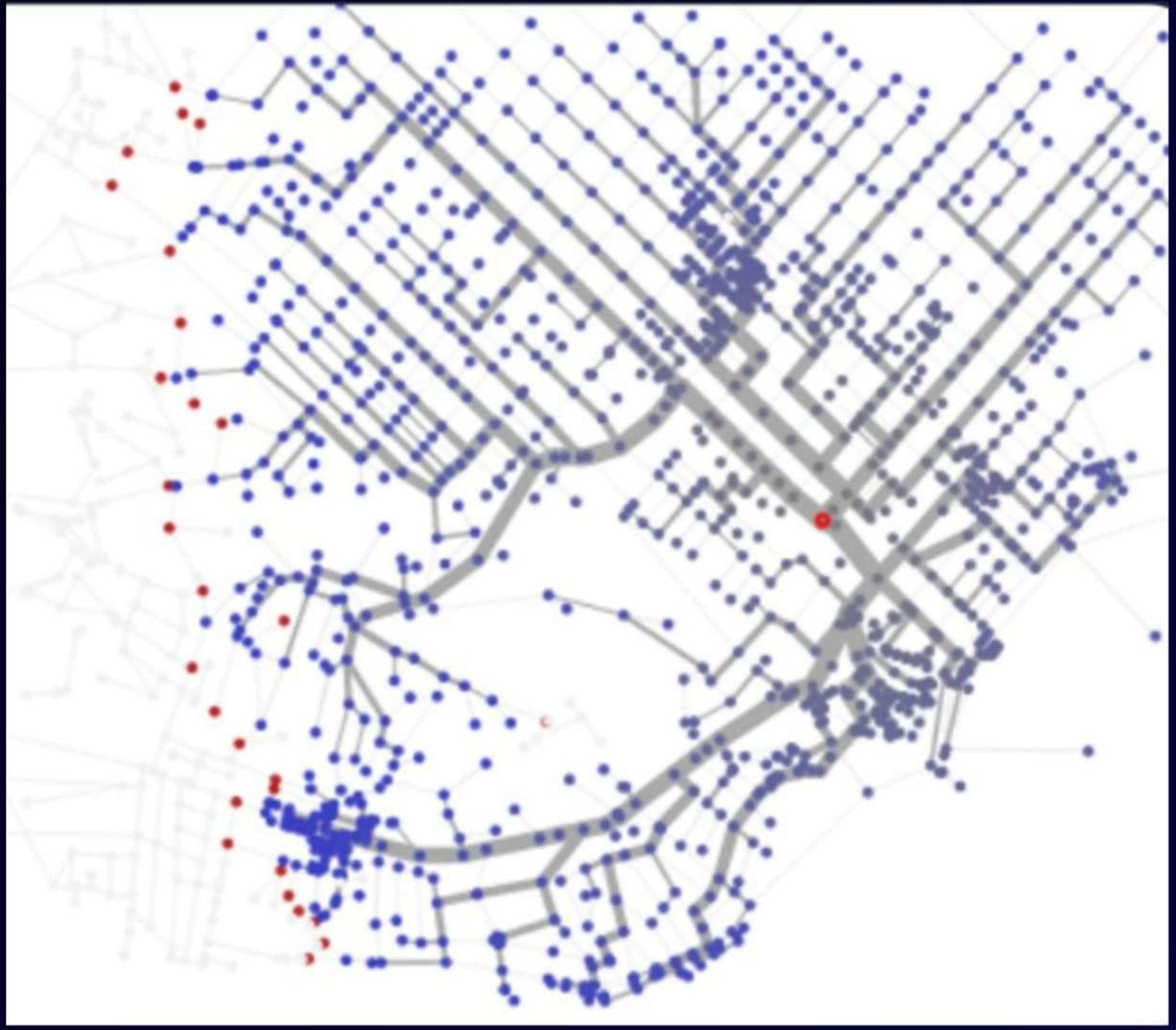
TASK C: CLUSTER DETECTION

Not every city may belong to the main network. Some cities or groups of cities may form isolated clusters. Find: Total number of connected components. Size of each component. Largest and smallest cluster.



TASK D: DIJKSTRA'S ALGORITHM

Select a source city and compute the shortest distance from that city to all other reachable cities.
Display: Shortest distance table. Unreachable cities (if any). City with maximum shortest-path distance from the source.



ALGORITHMIC PERFORMANCE

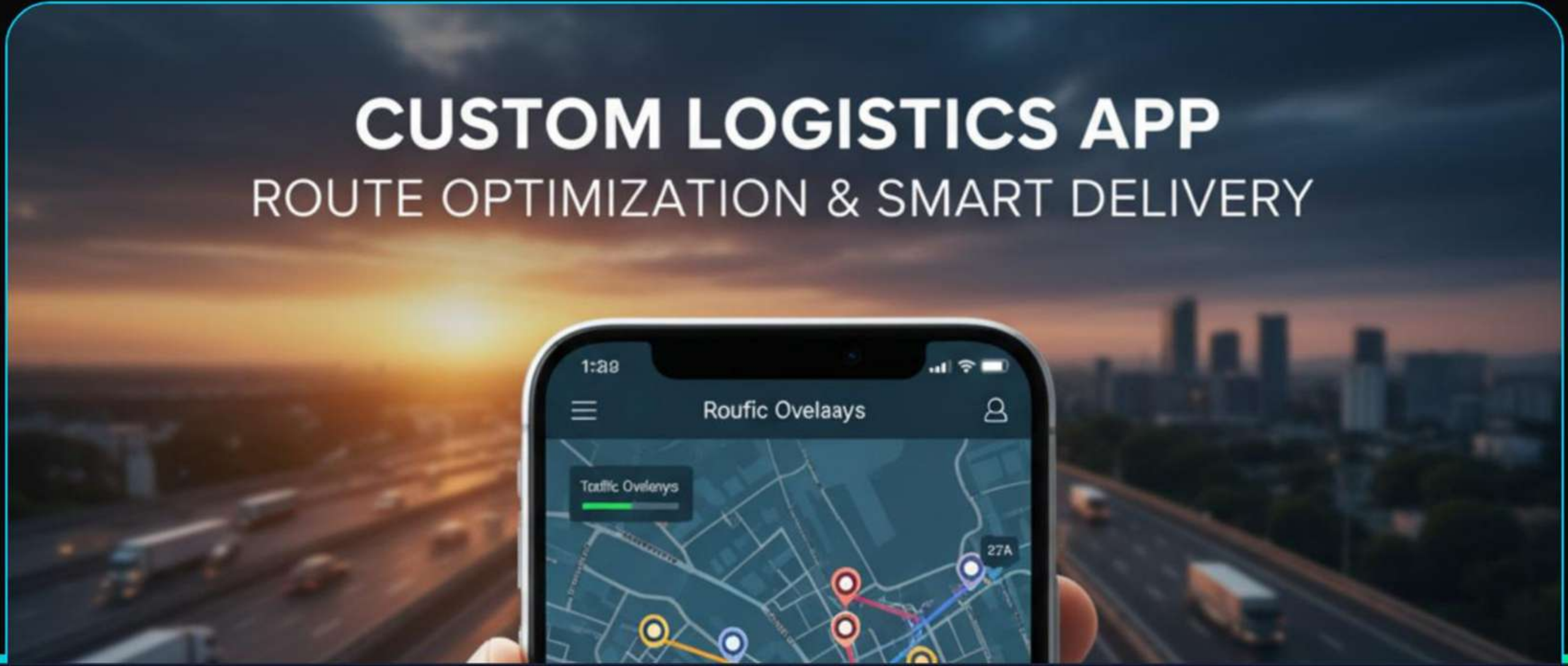
Algorithm	Primary Purpose	Time Complexity	Space Complexity
DFS / BFS	Connectivity & Clustering	O(V + E)	O(V)
Dijkstra	Shortest Path Optimization	O(E log V)	O(V)
Adjacency List	Graph Representation	N/A	O(V + E)

INDUSTRY APPLICATIONS

Logistics
Optimizing multi-stop delivery routes for fleets to minimize fuel and time.

Smart Maps
Real-time recalculation of routes based on dynamic road conditions.

Emergency
Ensuring first responders take the mathematically fastest path to critical incidents.



Thank You